Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Modeling Population Dynamics

1. Population Dynamics

Most species live in interaction with other species. For example, perhaps one species preys on another species, like foxes and rabbits. The given system of ODEs intends to predict future populations of rabbits and foxes over time, where R is the population (in thousands) of rabbits at time t and F is the population of foxes at time t (in years).

$$R' = 3R - \frac{7}{5}RF$$
$$F' = -F + \frac{4}{5}RF$$

a. For the rate of change R', population of rabbits R at time t, interpret each term of the equation in context.

b. For the rate of change F', population of foxes F at time t, interpret each term of the equation in context.

c. Determine all R-nullclines. Then, interpret them in context.

d. Determine all F-nullclines. Then, interpret them in context.

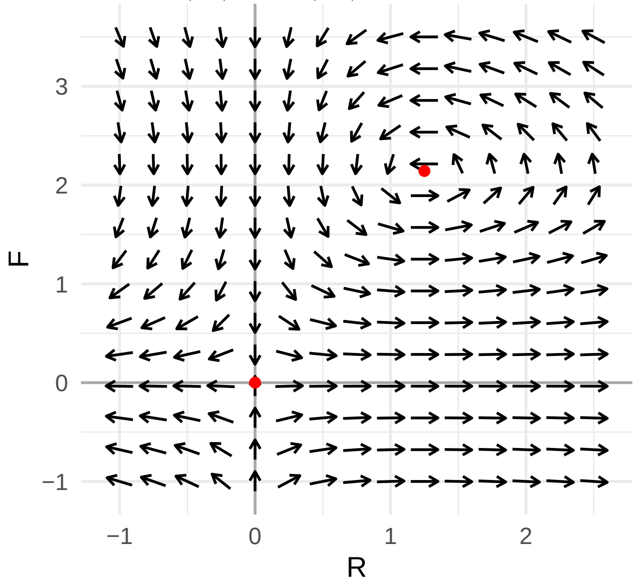
e. Use the nullclines determined in Parts (c) and (d) to compute the two equilibrium solutions, and name them (R_1, F_1) and (R_2, F_2).

2. Qualitative Analysis of Non-Linear Systems

Consider the same population dynamical system of two species from Problem 1. The rate of change in the rabbit R and fox F population at time t is modeled using a *homogeneous non-linear system of 1st-order ODEs*.

$$R' = 3R - \frac{7}{5}RF$$
$$F' = -F + \frac{4}{5}RF$$

a. Shown below is the vector field of the non-linear system. Draw the nullclines determined in Problems (1c) and (1d) on the phase plane.



b. Since this is a non-linear system, determining a general solution may not be possible, but we can linearize the system at each equilibrium to make solution behavior predictions.

   i. Determine the Jacobian matrix $J(R, F)$.

   ii. Determine the Jacobian matrix evaluated at each equilibrium. That is $J_1 = J(R_1, F_1)$ and $J_2 = J(R_2, F_2)$.

c. Determine the eigenvalues of J_1.

d. Determine the eigenvalues of J_2.

e. Use the eigenvalues determined in Parts (c) and (d) to describe the solution behaviors around each equilibrium.

f. Discuss how the eigenvalues for each equilibrium is represented in the vector field shown in Part (a).